Fiscal Year:	FY 2025	**Task Last Updated:**	FY 10/22/2024
PI Name:	Clark, Torin K. Ph.D.		
Project Title:	Detecting Pilot Spatial Disorientation to Trigger Active Countermeasures During Lunar Landing		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	80303-7813	**Congressional District:**	2
Comments:	NOTE: PI moved to University of Colorado after NSBRI Postdoctoral Fellowship concluded in late 2015 (Ed., 9/1/17)		
Project Type:	Ground	**Solicitation / Funding Source:**	2020-2021 HERO 80JSC020N0001-HHP, OMNIBUS3 Human Research Program: Human Health & Performance Appendix E; Omnibus3-Appendix F
Start Date:	12/23/2022	**End Date:**	12/22/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Holder, Sherrie Ph.D. (Charles Stark Draper Laboratory Inc) Endsley, Tristan Ph.D. (Charles Stark Draper Laboratory Inc) Vance, Eric Ph.D. (University of Colorado, Boulder) Dixon, Jordan (Charles Stark Draper Laboratory, Inc.)		
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Task Description:	November 2024 Update Per the Principal Investigator: in coordination with the NASA Human Research Program, the research team has successfully modified its original proposal to more extensively and precisely define the countermeasure intervention that would be triggered by its spatial disorientation algorithm. See the Task Progress section for additional information. [Ed., 11/6/24.] During transit in microgravity, astronauts will reinterpret neurovestibular stimuli, prior to initial exposure to partial gravity when landing on the Moon or Mars. This poses a risk of spatial disorientation and impaired manual control performance during piloted planetary landings. Here, we propose to develop, validate, and assess a system for detecting when astronauts may become disoriented in real-time, such that it can be used to trigger active countermeasures for piloted planetary landings. Our approach leverages a well-validated computational model for human spatial orientation, now applied to partial gravity planetary landings. Incorporating microgravity neurovestibular adaptation, the vehicle motions of each landing trajectory are processed in real-time by the computational model to detect pilot spatial disorientation. We will assess the system using a ground-based lunar landing analog, combining a gravity transition (3 Gx) with a motion-based planetary landing simulation. First, we will experimentally tune and re-validate the computational model for detecting spatial disorientation, accounting for the effects of the recent gravity transition. Then, using the high-fidelity Disorientation Research Device, we will assess the benefit of the active countermeasure triggering system. Critically, this approach of triggering manual control countermeasures only when they are needed (i.e., when the pilot is about to be disoriented) avoids the added burden on the pilot to continuously process additional sensory information or otherwise have increased workload. We aim to deliver a validated performance support tool for triggering active countermeasures for pilot spatial disorientation during manually controlled lunar landings.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We are developing a means to estimate a human's perception of self-orientation and the potential for spatial disorientation in real-time, in order to trigger an intervention to enhance performance and safety. In addition to the application of piloted lunar landing, our approach can be applied to pilots of terrestrial aircraft or other vehicles in which spatial orientation perception is critical. Finally, our enhancement of models for human spatial orientation perception applies to humans more generally, such as high performance performance athletes (e.g., gymnasts), critical operators (e.g., scuba divers), or patient populations (e.g., mTBI patients of older adults at risk for impaired balance).
Task Progress:	In this secondary year of the project, substantial technical, scientific, and administrative progress has been made. This project has involved extensive coordination with colleagues associated with the NASA Human Research Program, NASA Langley Research Center, and the Naval Aerospace Medical Research Laboratory - Dayton (NAMRU-D) in coordination for our experiment using the Disorientation Research Device (DRD). Through coordination with NASA Human Research Program personnel and managers, our team successfully modified our original proposal to more extensively and precisely define the countermeasure intervention we envision implementing and assessing, that would be triggered by our spatial disorientation algorithm. Through considering various alternatives, either that our team has previous experience with or those that have been proposed by others, we refined the potential alternatives. Our plan, the modification of the original proposal, was submitted to and approved by the NASA Human Research Program management team. The primary approach is for an adaptive display, which enhances the saliency (visual and auditory) of vehicle attitude and motion information in real-time based upon the spatial disorientation algorithm. Specifically, only when the algorithm deems the astronaut pilot is likely to be spatially disoriented does it automatically increase the saliency of critical information on the instrumentation display regarding vehicle attitude (roll and pitch) and motion to help the pilot reorient themselves in order to improve piloting performance and safety. When the algorithm estimates the pilot is not suffering from spatial disorientation, the nominal instrument display information (including landing point select, fuel remaining, terrain hazards, etc.) will be provided, avoiding an unnecessary burden upon the pilot. As such, we envision the adaptive display system will serve as a pilot aid, assisting the crewmember as needed. Completing this approval, we have exited the definition phase of our project. Next, we have coordinated with colleagues at NASA Langley for the detailed implementation of our countermeasure adaptive display. We have a site visit planned for mid-November to coordinate technical details of implementing our system (and other associated experimental needs) into their lunar-landing simulation and display environment. Finally, we have extensively coordinated with colleagues at NAMRU-D regarding plans for the capstone experiments using the DRD. This has included technical discussions of necessary instrumentation (e.g., accelerometers and inertial measurement units) for our planned experiments, as well as administrative coordination (e.g., developing Cooperative Research and Development Agreement / CRADA documentation). In addition to making progress in preparation for capstone experiments in the DRD, we completed our first set of preliminary experiments in our laboratory. First, we have developed a laboratory based capability for the hyper-Gx paradigm, planned for eventual use in the DRD, in a gravity transition analog. Participants spin on a centrifuge in hyper-Gx (i.e., the net force is “eye balls in”) for an extended period of time. When the centrifuge is spun down, the transition back from hyper-Gx tends to lead to impaired sensorimotor function and motion sickness, serving as an analog for astronaut gravity transitions. In our laboratory, the subject’s head is positioned 9 feet off-axis and spun to produce 2Gx for approximately 1 hour. In our first set of laboratory experiments, following the hyper-Gx exposure, subjects are transitioned to our Tilt Translation Sled (or TTS) and passively experience various tilts (and, in some cases, translation motions) and report their perception of tilt, using a common subjective haptic horizontal task. Beginning with roll tilt, perceptions are compared to those of subjects reporting in a control condition where they lay supine for 1 hour. This controls for the direction of the net gravito-inertial loading, but with only 1Gx, rather than 2Gx on the centrifuge. While perceptions of roll tilt are qualitatively similar, we found significant underestimation of roll tilt following the hyper-Gx paradigm, in comparison to that following laying Supine. Next, we will expand upon our results for roll tilt perception and study pitch tilt perception, where we anticipate the orientation of the hyper-Gx paradigm may induce a larger effect. We will incorporate the previous and upcoming findings into our computational model for spatial orientation perception and the associated algorithm for identifying spatial disorientation. In addition, we have made advancements to our human-rated motion device, the Tilt Translation Sled (or TTS), which we plan to use for preliminary investigations prior to the capstone assessments in the DRD. Specifically, we have implemented a manual control piloting mode that is analogous to lunar landing, whereby joystick deflections lead to a

	roll tilt motion, which is then coupled to a lateral acceleration in the same direction (i.e., tilt to the left leads to acceleration to the left). The subject is tasked with controlling translation motions via this coupled motion, similar to the manual control task during the final stages of lunar landing prior to touchdown. This control mode has been fully implemented and tested with pilot subjects in the loop. Finally, as an initial demonstration we have created a software implementation of our preliminary spatial disorientation algorithm on the TTS device. In real-time, the algorithm processes TTS motions (roll tilt and lateral translation) through our computational model for human spatial orientation perception and then computes a unidimensional metric of how disoriented the pilot is likely to be at that instant in time. The spatial disorientation metric is programmed on the TTS to trigger a change in the instrument display shown on a tablet in front of the participants.
Bibliography Type:	Description: (Last Updated: 11/28/2025)
Abstracts for Journals and Proceedings	Clark TK, Lonner TL, Dixon C, Holder S, Dixon JB, Endsley TC. "Detecting pilot spatial disorientation to trigger active countermeasures during lunar landing." 2024 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 13-16, 2024. Poster. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 13-16, 2024. , Feb-2024
Abstracts for Journals and Proceedings	Dixon C, Dixon J, Lonner TL, Endsley T, Clark TK. "Real-time detection of pilot spatial disorientation to trigger a pilot-aiding system." 94th Annual Scientific Meeting of the Aerospace Medical Association, Chicago, IL, May 5-9, 2024. Abstracts. 94th Annual Scientific Meeting of the Aerospace Medical Association, Chicago, IL, May 5-9, 2024. , May-2024
Papers from Meeting Proceedings	Lonner TL, Dixon C, Dixon JB, Endsley TC, Clark TK. "Countermeasure triggering for spatial disorientation experienced during piloted lunar landing." 2024 IEEE Aerospace Conference (AERO), Big Sky, MT, March 2-9, 2024. Proceedings from the 2024 IEEE Aerospace Conference (AERO), Big Sky, MT, March 2-9, 2024. http://dx.doi.org/10.1109/AERO58975.2024.10521322 , Mar-2024